LD12 Hybrid-Aware Density Scene Admission Specification

Exact support-atlas planning and mixed direct/FFT resource accounting

mdstats development specification

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Contents

Status and scope

This specification defines **LD12** for `mdstats 0.19.67a0`. It corrects the resource-admission mismatch between the LD0-R3 scene planner and the LD8-S3 production local-sparse executor.

The correction applies when

```
DensityOptimizationOptions(sparse_realization_mode="hybrid")
```

which is the production default. The explicit `sparse_realization_mode="ld7"` compatibility path retains its historical all-direct pair accounting.

LD12 does not alter:

- registered samples or their weights;
- cloud-in-cell deposition;
- the periodized Gaussian stencil;
- Gaussian bandwidth, grid interval, or tail tolerance;
- density normalization or HDR semantics;
- direct-tile or FFT-tile numerical kernels;
- browser mesh budgets.

The scientific scalar field is unchanged. LD12 changes which exact execution plan is admitted before allocation.

Problem statement

Before LD12, local-sparse Phase B built a streamed target union and reported

$$C_i = N_{\text{source},i} N_{\text{stencil},i}$$

as `kernel_pair_count` for field i. Phase C summed C_i over all fields and compared it with `max_density_kernel_pairs`.

That count is exact for the legacy all-direct scatter, but the production LD8-S3 executor partitions source nodes into compute tiles. A tile is evaluated either by direct stencil accumulation or by metric-aware overlap-add FFT convolution. Consequently, C_i is a valid count of mathematical source-stencil contributions, but it is not the number of direct pair operations executed by the hybrid algorithm.

The old planner could therefore reject a feasible scene when

$$\sum_i C_i > C_{\max}$$

although the actual direct work

$$D = \sum_i \sum_{t \in \mathcal{D}_i} N_{\text{source},t} N_{\text{stencil},i}$$

was below the direct-pair limit and the FFT tiles fit the wall-time and memory budgets.

Required planning order

For every production local-sparse candidate, Phase B must plan the same scientific and execution structures used during realization:

1. aggregate the globally weighted periodic CIC source;
2. construct the exact finite Gaussian stencil;
3. pack the CIC source into immutable source blocks;
4. construct or reuse the source-independent block-routing template;
5. build the exact finite-support target atlas;
6. partition the packed source into deterministic compute tiles;
7. select direct or FFT execution for each tile using the runtime-calibrated cost model;
8. record exact direct work, FFT work, target support, retained bytes, transient bytes, and estimated wall time;
9. submit those values to global Phase C admission.

No floating density values or mesh arrays are allocated in Phase B.

Data semantics

Exact mathematical contributions

For one field,

$$C_i = N_{\text{source},i} N_{\text{stencil},i}.$$

`exact_contribution_count` records C_i. It is a diagnostic and numerical identity check. It is not a direct-operation cap for the hybrid executor.

Direct-tile work

For the direct tile set $\mathcal{D}_i$,

$$D_i = \sum_{t \in \mathcal{D}_i} N_{\text{source},t} N_{\text{stencil},i}.$$

For a hybrid Phase-B record:

```
kernel_pair_count == direct_pair_count == D_i
kernel_pair_semantics == "actual_direct_tile_pairs"
```

Phase C compares

$$D_{\text{scene}} = \sum_i D_i$$

with `max_density_kernel_pairs`.

For the explicit LD7 compatibility path,

```
kernel_pair_count == C_i
kernel_pair_semantics == "all_direct_pairs"
```

because every contribution is evaluated directly.

FFT work

For each FFT tile $t \in \mathcal{F}_i$, let M_t be its padded FFT-node count. The selector estimates

$$T_{\text{FFT},t} = \alpha_{\text{FFT}} M_t \log_2 M_t + \beta_{\text{FFT}},$$

where the coefficients are derived from the current runtime calibration and cannot be relaxed below the measured conservative values by serialized options.

The field execution estimate is

$$T_i = \sum_{t \in \mathcal{D}_i} \alpha_{\text{direct}} D_t + \sum_{t \in \mathcal{F}_i} T_{\text{FFT},t}.$$

Phase C uses T_i directly. FFT work is not converted into fictitious direct pairs.

Phase-B record

The existing `DensityPhaseBFieldPlan` schema is retained. Its numerical fields have the following production-hybrid interpretation:

```
nonzero_node_count_upper = support_atlas.target_support_node_count
stored_value_count = support_atlas.target_support_node_count
stored_block_count = support_atlas.target_block_count
stencil_value_count = stencil.stencil_offset_count
kernel_pair_count = hybrid_plan.direct_pair_count
retained_bytes = hybrid_plan.packed_field_bytes_upper + retained_samples
transient_bytes_upper = conservative hybrid construction peak
```

The metadata records:

```
phase_b_execution_planner = ld8_s3_hybrid_exact_v1
phase_b_support_planner = ld8_s1_support_atlas_v1
exact_contribution_count
direct_pair_count
fft_padded_node_count
hybrid_compute_tile_count
hybrid_direct_tile_count
hybrid_fft_tile_count
hybrid_estimated_wall_seconds
hybrid_predicted_peak_bytes
hybrid_plan_identity
kernel_pair_semantics
```

`hybrid_plan_identity` is the SHA-256 identity of the exact immutable execution plan.

Phase-C scene admission

Let H be the hybrid fields, L the legacy all-direct sparse fields, and Q the dense fields. Phase C computes

$$D_{\text{scene}} = \sum_{i \in H} D_i + \sum_{i \in L} C_i.$$

The direct-operation guard is

$$D_{\text{scene}} \le D_{\max}.$$

The nominal total

$$C_{\text{scene}} = \sum_i C_i$$

is recorded as `total_exact_contributions` but is not compared with the direct pair limit for hybrid fields.

The preparation wall-time estimate is

$$T_{\text{scene}} = s\left[N_f\beta_f + \frac{N_s}{r_s} + \frac{N_k}{r_k} + \sum_{i\in H} T_i + \frac{\sum_{i\in L} C_i}{r_d} + \frac{N_{\text{dense}}}{r_{\text{dense}}}\right],$$

where s is the conservative safety multiplier, N_f is the field count, N_s is the total source-sample count, and N_k is the total retained stencil count.

A scene is rejected only when its selected mixed execution plan exceeds a memory, direct-work, or wall-time bound. The planner must not enlarge the grid interval or Gaussian bandwidth to make the scene fit.

Backend selection

`DensityBackendCandidateEstimate.estimated_work` uses the calibrated hybrid wall estimate for a production sparse candidate. Dense and sparse candidates therefore remain comparable without treating FFT tiles as all-direct work.

The field-local preference policy is unchanged:

- broad active support may prefer dense storage;
- localized support may prefer sparse storage;
- global Phase C may override field-local preferences to find a feasible scene;
- no candidate is considered feasible unless its own exact plan passes all runtime-derived limits.

Failure diagnostics

When the direct-pair limit fails, the error reports:

- actual direct/legacy pair count;
- configured `max_density_kernel_pairs`;
- hybrid field count;
- nominal exact contribution count;
- total FFT padded nodes.

When wall time fails, the error also reports the raw hybrid execution estimate. This distinguishes a genuinely expensive hybrid plan from the pre-LD12 false rejection.

Input constraints and edge cases

All-direct hybrid selection

If every hybrid tile selects direct execution, then $D_i = C_i$. LD12 reduces to the historical pair accounting and does not relax the guard.

All-FFT hybrid selection

If every tile selects FFT execution, then $D_i = 0$. The field is still bounded by FFT padded-node limits, peak memory, and calibrated wall time.

Mixed dense and sparse scene

Dense fields contribute dense-node work. Hybrid sparse fields contribute exact mixed execution time. Legacy sparse fields contribute all-direct pairs. The three paths are summed without double counting.

Explicit LD7 compatibility mode

LD7 target-union planning still enumerates streamed source-stencil pairs. Its pair count is real planning and execution work and remains subject to `max_density_kernel_pairs`.

Plan/realization mismatch

Realization recomputes immutable source, routing, atlas, and tile identities. The resulting target-node and target-block counts are compared with the Phase-B record. A mismatch is a planner defect, not a reason to alter numerical resolution.

Required tests

1. A hybrid field with `exact_contribution_count` above the direct-pair cap but `direct_pair_count` below it must pass Phase C.
2. A hybrid field with zero direct pairs but excessive FFT wall time must fail.
3. A forced local-sparse scene must record the exact hybrid planner and matching direct-pair semantics.
4. Direct and FFT tile counts must sum to the compute-tile count.
5. Scene metadata must separately record direct pairs, exact contributions, FFT padded nodes, and hybrid wall time.
6. The LD7 path must retain all-direct pair semantics.
7. Numerical density values must remain unchanged relative to the approved LD8 hybrid executor.

References

1. R. W. Hockney and J. W. Eastwood, *Computer Simulation Using Particles*, Taylor & Francis, 1988. Source of the standard cloud-in-cell particle-mesh assignment used before the project-specific packed support planning.
2. A. V. Oppenheim, R. W. Schafer, and J. R. Buck, *Discrete-Time Signal Processing*, 2nd ed., Prentice Hall, 1999. Source of the standard overlap-add convolution organization adapted by the LD8 tiled FFT executor.
3. `density_tiled_fft_ld8_s3_spec.md`, project specification for the project-specific metric-aware periodic direct/FFT tile selector.
4. `density_support_atlas_ld8_s1_spec.md`, project specification for exact packed finite-support atlas construction.